(i) Let $t = 1 - x,\ dx = -dt,\ \begin{array}{c|cc} x & 0 & 1 \\ \hline t & 1 & 0 \end{array}$

$$\begin{aligned}\int_0^1 x^{n-1}(1-x)^n\,dx &= \int_1^0 (1-t)^{n-1}t^n(-dt)\\ &= \int_0^1 t^n(1-t)^{n-1}\,dt\\ &= \int_0^1 x^n(1-x)^{n-1}\,dx\end{aligned}$$

$$\begin{aligned}I_{n-1} &= \int_0^1 x^{n-1}(1-x)^{n-1}\,dx\\ &= \int_0^1 x^{n-1}(1-x)^{n-1}\big(x+1-x\big)\,dx\\ &= \int_0^1 x^n(1-x)^{n-1}\,dx + \int_0^1 x^{n-1}(1-x)^n\,dx\\ &= 2\int_0^1 x^{n-1}(1-x)^n\,dx\end{aligned}$$

$$u = x^n,\ dv = (1-x)^n\,dx,\ du = nx^{n-1}\,dx,\ v = -\frac{(1-x)^{n+1}}{n+1}$$

$$\begin{aligned}I_n &= \left[-\frac{x^n(1-x)^{n+1}}{n+1}\right]_0^1 + \frac{n}{n+1}\int_0^1 x^{n-1}(1-x)^{n+1}\,dx\\ &= \frac{n}{n+1}\int_0^1 x^{n-1}(1-x)^{n+1}\,dx\end{aligned}$$

Let $A = \displaystyle\int_0^1 x^{n-1}(1-x)^n\,dx = \frac{1}{2}I_{n-1}$

$$\int_0^1 x^{n-1}(1-x)^{n+1}\,dx = \int_0^1 x^{n-1}(1-x)^n(1-x)\,dx = A - I_n$$

$$I_n = \frac{n}{n+1}(A - I_n)$$

$$I_n = \frac{n}{n+1}\left(\frac{1}{2}I_{n-1} - I_n\right)$$

$$\left(1 + \frac{n}{n+1}\right)I_n = \frac{n}{2(n+1)}I_{n-1}$$

$$\therefore I_n = \frac{n}{2(2n+1)}I_{n-1}$$

(ii)

$$I_0 = \int_0^1 1\,dx = 1$$

$$I_n = \frac{n}{2(2n+1)} I_{n-1}$$

$$\begin{aligned} I_n &= \frac{n}{2(2n+1)} \cdot \frac{n-1}{2(2n-1)} \cdots \frac{1}{2\cdot 3} \cdot I_0 \\ &= \frac{n!}{2^n \cdot 3 \cdot 5 \cdots (2n+1)} \\ &= \frac{n!}{2^n \,.\, \frac{(2n+1)!}{2^n n!}} = \frac{(n!)^2}{(2n+1)!} \end{aligned}$$

(iii) When $x = \sin^2\theta,\ dx = 2\sin\theta\cos\theta\,d\theta,\ x = 0 \Rightarrow \theta = 0,\ x = 1 \Rightarrow \theta = \dfrac{\pi}{2}$

$$\begin{aligned} I_{\frac{1}{2}} &= \int_0^1 x^{1/2}(1-x)^{1/2}\,dx \\ &= \int_0^{\pi/2} \sin\theta\cos\theta \cdot 2\sin\theta\cos\theta\,d\theta \\ &= 2\int_0^{\pi/2} \sin^2\theta\cos^2\theta\,d\theta \\ &= \frac{1}{2}\int_0^{\pi/2} \sin^2 2\theta\,d\theta \\ &= \frac{1}{4}\int_0^{\pi/2} (1-\cos 4\theta)\,d\theta \\ &= \frac{1}{4}\left[\theta - \frac{1}{4}\sin 4\theta\right]_0^{\pi/2} \\ &= \frac{\pi}{8} \end{aligned}$$

$$I_{\frac{3}{2}} = \frac{\frac{3}{2}}{2(2\cdot\frac{3}{2}+1)} I_{\frac{1}{2}} = \frac{3}{16}\cdot\frac{\pi}{8} = \frac{3\pi}{128}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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